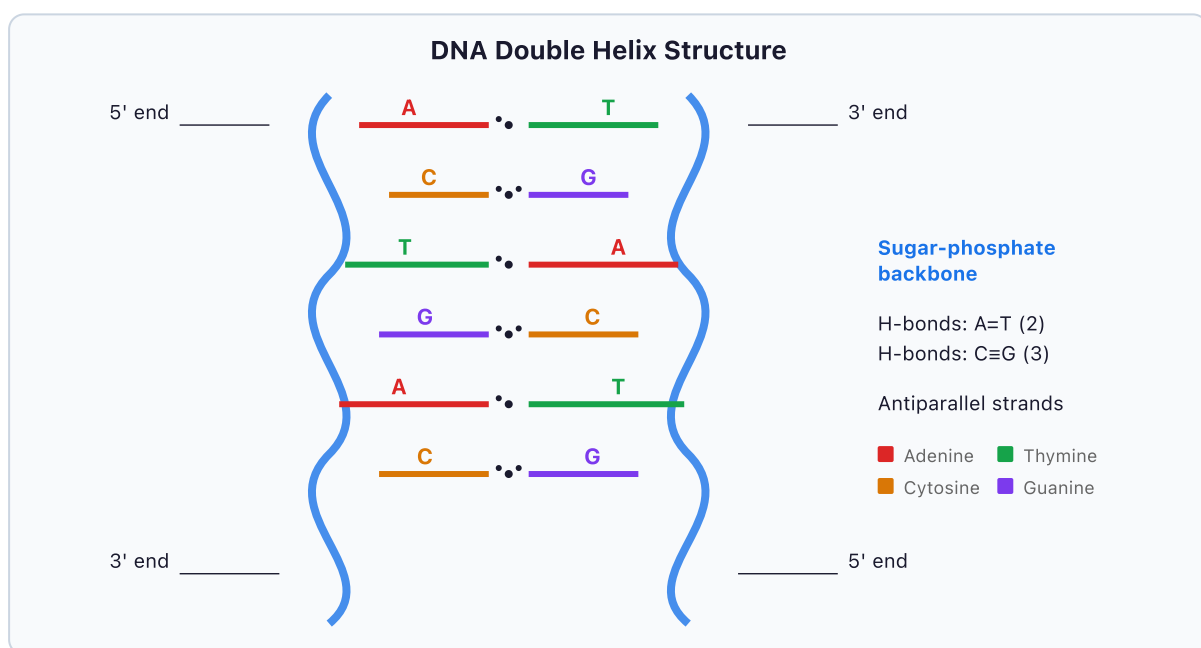


S2 — Genes and Health

2.1 DNA Structure and the Genetic Code

DNA Structure

DNA (deoxyribonucleic acid): A double-stranded polynucleotide that carries the genetic information of an organism. Each nucleotide consists of a deoxyribose sugar, a phosphate group, and one of four nitrogenous bases: adenine (A), thymine (T), cytosine (C), or guanine (G).



The two polynucleotide strands are held together by hydrogen bonds between complementary base pairs: adenine pairs with thymine (two hydrogen bonds) and cytosine pairs with guanine (three hydrogen bonds). The strands run antiparallel (one runs 5' to 3', the other 3' to 5') and are twisted into a right-handed double helix with approximately 10 base pairs per turn. The sugar-phosphate backbone provides structural stability (covalent phosphodiester bonds are strong) while the bases face inward carrying the genetic information. The two strands are complementary, meaning the sequence of one strand determines the sequence of the other — this is crucial for DNA replication and transcription.

Nucleotide structure: Each nucleotide is formed by a condensation reaction joining a pentose sugar (deoxyribose in DNA, ribose in RNA), a phosphate group, and a nitrogenous base. Nucleotides are joined by phosphodiester bonds between the 3' carbon of one sugar and the 5' phosphate of the next, creating a directional backbone. Purines (adenine, guanine) have a double-ring structure; pyrimidines (thymine, cytosine) have a single ring. A purine always pairs with a pyrimidine, keeping the helix width constant.

Chargaff's rules: In any DNA molecule, the amount of adenine equals thymine, and the amount of cytosine equals guanine ($A = T$, $C = G$). This was key evidence supporting the Watson-Crick model of complementary base pairing. The ratio of A+T to C+G varies between species but is constant within a species.

Genes and the Genetic Code

Gene: A sequence of DNA nucleotides that codes for a functional polypeptide (or functional RNA molecule). Genes are located at specific positions (loci) on chromosomes.

The genetic code has the following features:

- **Triplet:** Each codon consists of three consecutive bases coding for one amino acid.
- **Degenerate (redundant):** Most amino acids are coded for by more than one codon (there are 64 possible codons but only 20 amino acids).
- **Non-overlapping:** Each base is read only once, as part of only one codon.
- **Universal:** The same codons code for the same amino acids in almost all organisms (evidence of common ancestry).
- **Contains start and stop codons:** AUG is the start codon (also codes for methionine); UAA, UAG, and UGA are stop codons.

DNA Replication (Semi-Conservative)

DNA replication occurs during the S phase of the cell cycle and is described as **semi-conservative** because each new DNA molecule consists of one original (parent) strand and one newly synthesised strand. The process:

1. **Unwinding:** The enzyme **DNA helicase** unwinds the double helix and breaks the hydrogen bonds between complementary base pairs, separating the two strands and creating a replication fork.

2. **Priming:** RNA primase synthesises a short RNA primer on each template strand, providing a free 3'-OH group for DNA polymerase to begin synthesis.
3. **Elongation: DNA polymerase** reads the template strand in the 3' to 5' direction and synthesises the new complementary strand in the 5' to 3' direction, adding free DNA nucleotides by complementary base pairing (A with T, C with G). The leading strand is synthesised continuously; the lagging strand is synthesised in short Okazaki fragments.
4. **Joining: DNA ligase** joins the Okazaki fragments on the lagging strand by forming phosphodiester bonds, creating a continuous strand.
5. **Proofreading:** DNA polymerase has 3' to 5' exonuclease activity, allowing it to remove incorrectly paired nucleotides and replace them, giving an error rate of approximately 1 in 10^9 bases. Errors that escape proofreading (a wrong base inserted, or bases added or lost) become permanent **mutations**: changes in the base sequence that are passed on to the daughter cells when the DNA is replicated again.

Evidence for semi-conservative replication: The Meselson-Stahl experiment (1958) used *E. coli* grown in heavy nitrogen (^{15}N) medium, then transferred to light nitrogen (^{14}N) medium. After one round of replication, all DNA had intermediate density (one heavy and one light strand). After two rounds, half had intermediate density and half had light density. This conclusively supported semi-conservative replication over conservative or dispersive models.

Exam Tip: When describing DNA replication, always name the enzymes (helicase, DNA polymerase, ligase) and state their specific roles. A common mistake is saying DNA polymerase "moves along" the template — be precise: it reads 3'→5' and synthesises 5'→3'. Remember that replication is bidirectional (two replication forks move in opposite directions from the origin).

RNA Structure

RNA differs from DNA in several ways: it is usually single-stranded (though it can fold into complex secondary structures via intramolecular base pairing), contains ribose sugar instead of deoxyribose (ribose has an -OH group on the 2' carbon), uses uracil (U) instead of thymine, and is generally shorter and less stable. The main types are:

- **Messenger RNA (mRNA):** Carries the genetic code from DNA in the nucleus to ribosomes in the cytoplasm. Its sequence of codons determines the amino acid sequence of the protein.

- **Transfer RNA (tRNA):** Small, clover-leaf shaped molecules that carry specific amino acids to the ribosome during translation. Each tRNA has an anticodon complementary to an mRNA codon and an amino acid attachment site at the 3' end.
- **Ribosomal RNA (rRNA):** A structural and catalytic component of ribosomes, making up about 60% of ribosome mass. rRNA has peptidyl transferase activity, catalysing peptide bond formation.

2.2 Protein Synthesis

Transcription

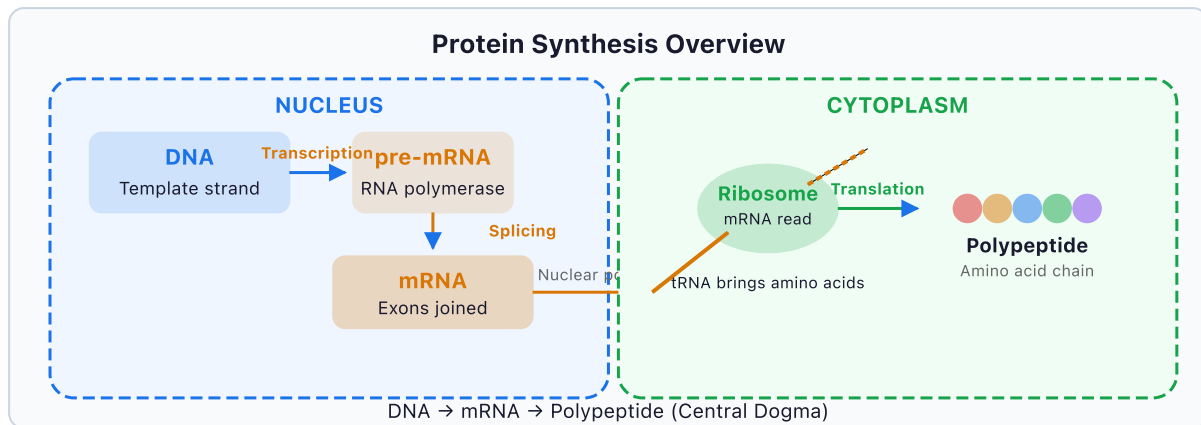
Transcription occurs in the nucleus. The enzyme RNA polymerase binds to a promoter region on the DNA template strand and unwinds the double helix. RNA polymerase moves along the template strand in the 3' to 5' direction, synthesising a complementary mRNA strand in the 5' to 3' direction using free RNA nucleotides. Hydrogen bonds between DNA strands are broken ahead of the enzyme and reform behind it. When RNA polymerase reaches a terminator sequence, transcription stops and the pre-mRNA is released.

In eukaryotes, the pre-mRNA undergoes processing: introns (non-coding sequences) are spliced out and exons (coding sequences) are joined together. A 5' cap and 3' poly-A tail are added for stability and to aid export from the nucleus through nuclear pores.

Translation

Translation occurs at ribosomes in the cytoplasm. The mature mRNA binds to a ribosome. Transfer RNA (tRNA) molecules, each carrying a specific amino acid, have anticodons that are complementary to mRNA codons. The ribosome moves along the mRNA, reading codons sequentially. When a tRNA anticodon pairs with the mRNA codon in the ribosome's A site, the amino acid is joined to the growing polypeptide chain by a peptide bond (catalysed by peptidyl transferase in the ribosome). The tRNA then exits from the E site. Translation continues until a stop codon is reached, at which point a release factor binds and the polypeptide is released.

Exam Tip: Be precise about which strand is the template. The template strand is read 3' to 5'; the mRNA produced is complementary to it. The coding (sense) strand has the same sequence as the mRNA (with T instead of U). Always specify the direction of synthesis.



Post-Translational Modification

After translation, the polypeptide folds into its secondary structure (alpha helices and beta pleated sheets held by hydrogen bonds), tertiary structure (further folding stabilised by ionic bonds, disulfide bridges, hydrophobic interactions, and hydrogen bonds), and potentially quaternary structure (multiple polypeptide subunits). Some proteins require the addition of non-protein groups such as haem (in haemoglobin) or carbohydrate chains (glycoproteins).

Amino Acids and Peptide Bonds

Every amino acid has the same basic structure: a central carbon atom bonded to an **amino group** ($-\text{NH}_2$), a **carboxyl group** ($-\text{COOH}$), a hydrogen atom and a variable **R group**. The R group is different in each of the 20 amino acids. Two amino acids join in a **condensation reaction** between the carboxyl group of one and the amino group of the other. This forms a **peptide bond** and releases a molecule of water, giving a dipeptide. Many amino acids joined in this way form a polypeptide. Hydrolysis reactions break peptide bonds.

Protein Structure and Properties

The **primary structure** (the sequence of amino acids) determines where hydrogen bonds, ionic bonds, disulfide bridges and hydrophobic interactions form between R groups. It therefore determines how the chain folds into its three-dimensional shape, and so the protein's properties. Changing a single amino acid can change the shape of the protein and stop it working.

- **Globular proteins** are compact and roughly spherical. Hydrophilic R groups face outwards and hydrophobic R groups face inwards, so globular proteins are soluble and suit metabolic roles (e.g. enzymes, transport). **Haemoglobin** is a globular protein made of four polypeptide chains (two α and two β).

two β), each with a haem group containing an iron ion (Fe^{2+}) that binds oxygen. It is soluble, so it can be carried in red blood cells, and it transports oxygen.

- **Fibrous proteins** are long, insoluble chains with structural roles. **Collagen** consists of three polypeptide chains wound around each other in a triple helix held together by hydrogen bonds. Every third amino acid is glycine, the smallest amino acid, which lets the chains pack tightly. Covalent cross-links between neighbouring collagen molecules form strong fibres. Collagen has high tensile strength and is found in artery walls, tendons, skin and bone.

Enzymes

Enzymes are globular proteins that act as **biological catalysts**. They speed up reactions without being used up by **lowering the activation energy** needed for the reaction. Each enzyme has an **active site** whose specific three-dimensional shape is determined by the enzyme's tertiary structure. Only a substrate with a complementary shape can bind, which makes enzymes **specific**. The substrate binds to form an **enzyme–substrate complex**. In the **induced fit** model, the active site changes shape slightly as the substrate binds, which strains the bonds in the substrate or holds reacting molecules close together. This lowers the activation energy. The products are then released and the enzyme can be used again.

Intracellular enzymes catalyse reactions inside cells (e.g. DNA polymerase, and the enzymes of respiration). **Extracellular enzymes** are secreted by cells, by exocytosis, and catalyse reactions outside them (e.g. the digestive enzymes amylase and trypsin, released from the pancreas into the small intestine).

Enzyme and substrate concentration: The **initial rate** of reaction is measured at the start, before substrate is used up. It is found by drawing a tangent to the product–time curve at time zero and calculating its gradient. As substrate concentration increases, the initial rate increases because more enzyme–substrate complexes form. The rate then levels off when all the active sites are occupied, and enzyme concentration becomes the limiting factor. When substrate is in excess, the initial rate is directly proportional to enzyme concentration.

Core Practical 4: Investigate the effect of enzyme and substrate concentrations on the initial rates of reactions

An example uses the enzyme catalase (from yeast suspension or potato/celery extract), which breaks down hydrogen peroxide into water and oxygen. Collect the oxygen over water in an inverted measuring cylinder (or in a gas syringe) and record the volume every 10 seconds. Vary the hydrogen peroxide

concentration by dilution (or vary the enzyme concentration), keeping everything else constant: temperature (water bath), pH (buffer), and the volumes of enzyme and substrate. Plot the volume of oxygen against time, draw a tangent at the origin to find each initial rate, then plot initial rate against concentration. (Trypsin acting on milk protein, followed with a colorimeter, is an alternative.)

2.3 Gene Mutations and Genetic Disease

Types of Gene Mutation

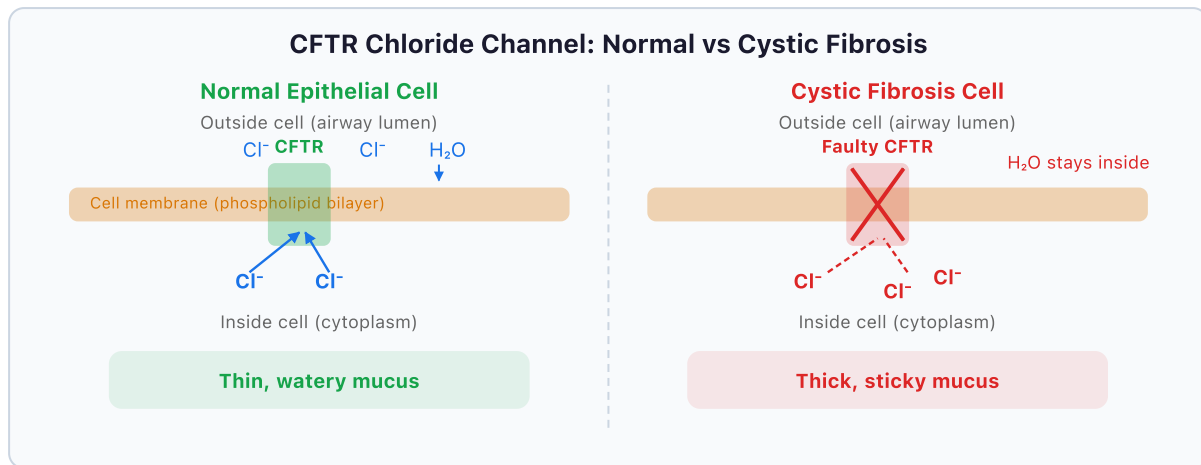
Gene mutation: A change in the nucleotide sequence of DNA. This may be a substitution (one base replaced by another), insertion (an extra base added), or deletion (a base removed). Insertions and deletions of a number of bases that is not a multiple of three cause a frameshift, altering every codon downstream of the mutation.

The consequences of mutations depend on their type and location:

- **Silent mutation:** The new codon still codes for the same amino acid (due to degeneracy of the code). No effect on the protein.
- **Missense mutation:** The new codon codes for a different amino acid. May or may not affect protein function depending on the nature of the change and its position.
- **Nonsense mutation:** The new codon is a stop codon, resulting in a truncated (shortened) and usually non-functional protein.
- **Frameshift mutation:** An insertion or deletion shifts the reading frame, changing all downstream amino acids. Almost always produces a non-functional protein.

Cystic Fibrosis

Cystic fibrosis (CF): An autosomal recessive genetic disorder caused by mutations in the CFTR gene on chromosome 7. The CFTR protein is a chloride ion channel in epithelial cell membranes. When it is absent or non-functional, chloride ions cannot be secreted out of cells, water does not follow by osmosis, and mucus becomes thick and sticky.



The most common mutation is ΔF508 — a deletion of three nucleotides that removes phenylalanine at position 508. This causes the CFTR protein to misfold so it is degraded before reaching the cell membrane. Over 2000 different mutations have been identified in the CFTR gene.

Symptoms of CF include:

- **Lungs:** Thick mucus clogs airways, reducing gas exchange surface area and trapping bacteria, leading to recurrent infections and progressive lung damage.
- **Pancreas:** Mucus blocks pancreatic ducts, preventing digestive enzymes from reaching the small intestine, causing malabsorption and malnutrition.
- **Reproductive system:** Thick cervical mucus in females reduces fertility; males often lack the vas deferens.
- **Sweat glands:** Produce abnormally salty sweat (used as a diagnostic test).

Genetic Inheritance of CF

CF is autosomal recessive, meaning an individual must inherit two copies of the faulty allele (be homozygous recessive, ff) to show the disease. Carriers (heterozygous, Ff) have one functional copy and one faulty copy — they produce enough functional CFTR protein to be unaffected but can pass the allele to offspring. If two carriers have a child, there is a 1 in 4 (25%) chance the child will have CF, a 1 in 2 (50%) chance of being a carrier, and a 1 in 4 chance of being homozygous dominant.

Genetic Cross: Two CF Carriers (Ff x Ff)Father: **Ff** (carrier)Mother: **Ff** (carrier)

Gametes: F or f

Gametes: F or f

	F	f
F	FF Unaffected	Ff Carrier
f	Ff Carrier	ff Has CF

25% **FF**50% **Ff**25% **ff**Phenotype ratio = **3 unaffected : 1 CF**

F = normal CFTR allele (dominant) | f = faulty CFTR allele (recessive)

Exam Tip: In genetics questions, always define your allele symbols, draw a genetic diagram showing parental genotypes, gametes, a Punnett square, and offspring ratios. State the probability as a fraction, percentage, or ratio.

Genetic Terms

Term	Meaning
Gene	A sequence of bases on a DNA molecule that codes for a sequence of amino acids in a polypeptide chain
Allele	An alternative form of a gene
Genotype	The alleles an organism has for a particular gene (e.g. Ff)
Phenotype	The observable characteristics of an organism, which result from its genotype and the environment
Dominant	An allele that is expressed in the phenotype whenever it is present
Recessive	An allele that is only expressed in the phenotype when two copies are present (no dominant allele)

Incomplete dominance	Neither allele is completely dominant, so the heterozygote has an intermediate phenotype (e.g. red × white snapdragons give pink offspring; crossing two pink plants gives a 1 red : 2 pink : 1 white ratio)
Homozygote	An individual with two identical alleles of a gene (e.g. FF or ff)
Heterozygote	An individual with two different alleles of a gene (e.g. Ff)

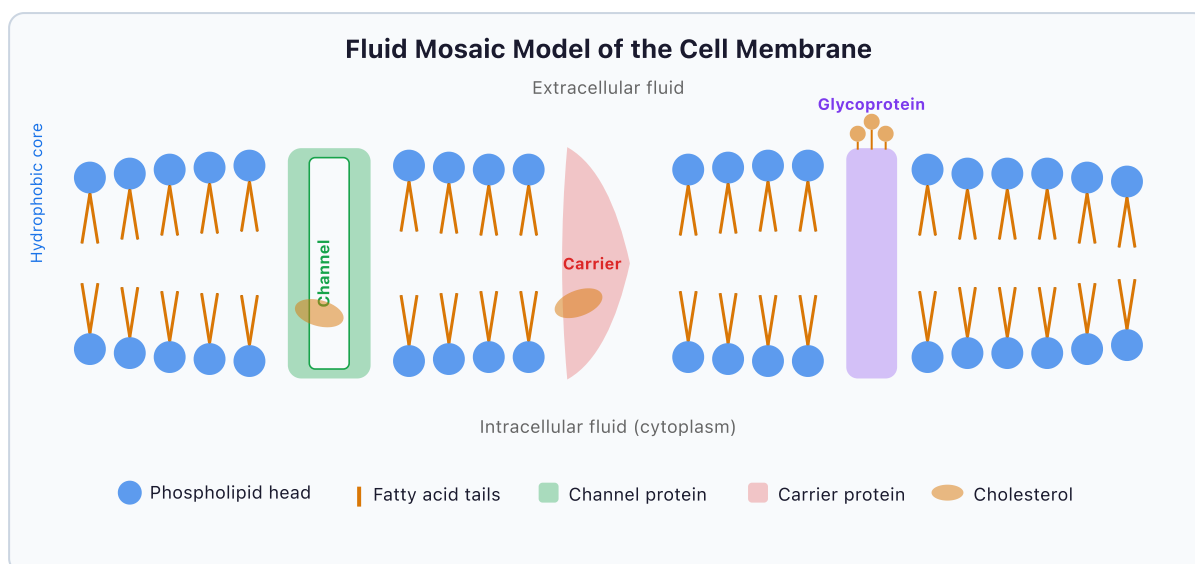
Pedigree Diagrams

A genetic pedigree diagram shows how a characteristic is inherited in a family. Squares are males and circles are females. Shaded symbols are affected individuals; carriers are sometimes shown half-shaded. A horizontal line joins parents, and vertical lines lead down to their children. If two unaffected parents have an affected child, the condition must be caused by a recessive allele; for an autosomal condition such as CF, both parents must be heterozygous carriers. Work out the genotypes of the affected (homozygous recessive) individuals first, then deduce the genotypes of the others.

2.4 Cell Membranes

Fluid Mosaic Model

Fluid mosaic model: The accepted model of membrane structure in which a phospholipid bilayer forms a fluid matrix, with proteins (integral and peripheral), cholesterol, and glycolipids/glycoproteins scattered throughout in a mosaic pattern.



The phospholipid bilayer is the fundamental structure. Each phospholipid has a hydrophilic phosphate head and two hydrophobic fatty acid tails. In aqueous solution they spontaneously form a bilayer with heads facing outward and tails facing inward, creating a hydrophobic barrier to water-soluble substances.

Key components of the cell membrane:

Component	Function
Phospholipid bilayer	Forms the basic barrier; allows fat-soluble substances through; fluid structure allows membrane flexibility
Cholesterol	Regulates membrane fluidity — prevents phospholipids from packing too closely at low temperatures or moving too freely at high temperatures
Channel proteins	Provide hydrophilic pores for specific ions/polar molecules to pass through (passive transport)
Carrier proteins	Change shape to transport specific molecules across the membrane (facilitated diffusion or active transport)
Glycoproteins	Cell signalling, cell recognition, receptors for hormones and drugs
Glycolipids	Cell recognition and stability

Models of Membrane Structure

The fluid mosaic model is an interpretation of experimental data, and it replaced earlier models as new evidence appeared. Lipid-soluble substances enter cells faster than water-soluble ones, which suggested that membranes contain lipid. Lipid extracted from red blood cell membranes covers about twice the surface area of the cells, which suggested a bilayer. The **Davson–Danielli model** (1935) proposed a lipid bilayer sandwiched between two continuous layers of protein, and early electron micrographs showing a three-layered (“railway track”) membrane appeared to support it. Later evidence did not fit. Freeze-fracture electron microscopy showed proteins embedded within and spanning the bilayer, and when cells with fluorescently labelled membrane proteins were fused, the proteins quickly mixed, showing they can move. **Singer and Nicolson** therefore proposed the fluid mosaic model (1972). This shows how scientific models change when new data cannot be explained by the existing model.

Core Practical 3: Investigate membrane structure, including the effect of alcohol concentration or temperature on membrane permeability

Beetroot cells contain a red pigment (betalain) in their vacuoles, which leaks out only if the membranes are damaged. Cut equal-sized pieces of beetroot with a cork borer and rinse them in distilled water to remove pigment from cells damaged by cutting. Place the pieces in equal volumes of water at a range of temperatures (in water baths), or in a range of ethanol concentrations, for a fixed time. Remove the beetroot and measure the absorbance of the surrounding liquid with a colorimeter (blue-green filter). Higher absorbance means more pigment has leaked out, so the membranes are more permeable. As temperature rises, phospholipids gain kinetic energy and the membrane becomes more fluid, and at high temperatures membrane proteins denature, leaving gaps. Ethanol, a non-polar solvent, dissolves and disrupts the phospholipid bilayer, so permeability increases with ethanol concentration.

Transport Across Membranes

Diffusion: The net movement of molecules or ions from a region of higher concentration to a region of lower concentration, down a concentration gradient. It is passive (requires no ATP) and continues until equilibrium is reached. Rate is affected by concentration gradient, temperature, surface area, and thickness of the membrane.

Facilitated diffusion: Diffusion of molecules or ions across a membrane via specific channel or carrier proteins. Still passive and moves down the concentration gradient, but allows hydrophilic or charged particles that cannot cross the phospholipid bilayer to pass through.

Osmosis: The net movement of free water molecules through a partially permeable membrane, from a solution with a lower solute concentration (more free water molecules) to a solution with a higher solute

concentration (fewer free water molecules). Water molecules that are attracted to (clustered around) solute molecules are not free to move, so the more concentrated solution has fewer free water molecules. 9BN0 describes osmosis in these terms and does not require water potential; other courses describe the same movement as being from a higher to a lower water potential.

Active transport: The movement of molecules or ions against their concentration gradient, from low concentration to high concentration, using ATP as an energy source. Requires carrier proteins that change shape when ATP is hydrolysed. Example: sodium-potassium pump (Na^+/K^+ -ATPase) pumps 3 Na^+ out and 2 K^+ in per ATP molecule, maintaining the electrochemical gradient essential for nerve impulse transmission. Cells that carry out extensive active transport (e.g., root hair cells, kidney tubule cells, intestinal epithelial cells) have many mitochondria to supply the required ATP.

Endocytosis and exocytosis: Large molecules and particles that cannot cross the membrane via proteins are transported in vesicles. **Endocytosis** is the uptake of material into a cell by invagination of the membrane to form a vesicle (phagocytosis for solids, pinocytosis for liquids). **Exocytosis** is the reverse: vesicles fuse with the cell membrane to release their contents outside the cell (e.g., secretion of hormones, neurotransmitters, digestive enzymes). Both processes require ATP.

Exam Tip: When comparing transport mechanisms, organise your answer around key criteria: does it require energy (ATP)? Does it require a protein? Does it move down or against a concentration gradient? Is it specific to certain molecules? Diffusion: no energy, no protein, down gradient. Facilitated diffusion: no energy, protein required, down gradient. Active transport: energy required, protein required, against gradient. Osmosis: no energy, no protein (but requires partially permeable membrane), water only.

2.5 Gas Exchange in the Lungs

Structure of the Gas Exchange System

The human gas exchange system consists of the trachea (reinforced with C-shaped cartilage rings), bronchi, bronchioles, and alveoli. The trachea and bronchi are lined with ciliated epithelium and goblet cells that produce mucus to trap pathogens and particulates; cilia beat rhythmically to move mucus up to the throat (the mucociliary escalator).

Alveoli and Gas Exchange

Gas exchange occurs across the alveolar surface by diffusion. Oxygen diffuses from the alveolar air into the blood in pulmonary capillaries, while carbon dioxide diffuses in the opposite direction. The alveoli are highly adapted for efficient gas exchange:

- **Large surface area:** ~70 m² provided by approximately 300 million alveoli.
- **Thin diffusion distance:** Alveolar wall is one cell thick, capillary wall is one cell thick — total diffusion distance is approximately 0.5 µm.
- **Rich blood supply:** Dense capillary network maintains a steep concentration gradient by continuously removing oxygen and delivering carbon dioxide.
- **Ventilation:** Breathing movements maintain a steep concentration gradient by replacing alveolar air with fresh atmospheric air.
- **Moist surface:** Gases dissolve in the thin layer of moisture lining the alveoli before diffusing across the membrane.

Fick's law of diffusion: The rate of diffusion is proportional to (surface area × concentration difference) / thickness of the exchange surface. This summarises the adaptations of gas exchange surfaces.

$$\text{Rate of diffusion} \propto (\text{surface area} \times \Delta\text{concentration}) / \text{distance}$$

Gas Exchange and Cystic Fibrosis

In CF patients, thick sticky mucus accumulates in the airways because defective CFTR channels cannot secrete chloride ions, so water does not follow by osmosis to hydrate the mucus. This thick mucus: blocks bronchioles, reducing ventilation of alveoli; is difficult for cilia to move, so the mucociliary escalator is impaired; traps bacteria (especially *Pseudomonas aeruginosa* and *Staphylococcus aureus*), leading to chronic infections and inflammation. Over time, repeated infections cause scarring and fibrosis of lung tissue, reducing the surface area for gas exchange and ultimately leading to respiratory failure.

CF Treatments

Current treatments manage symptoms rather than curing the underlying genetic defect:

- **Physiotherapy (chest percussion/postural drainage):** Loosens and drains thick mucus from the lungs.
- **Mucolytics (e.g., DNase/dornase alfa):** Enzyme that breaks down DNA released from dead neutrophils in the mucus, reducing its viscosity.
- **Antibiotics:** Used prophylactically and to treat acute lung infections. Nebulised antibiotics deliver drugs directly to the lungs.
- **Pancreatic enzyme supplements:** Taken with meals to replace the digestive enzymes that cannot reach the small intestine due to blocked pancreatic ducts.
- **Bronchodilators:** Widen airways to improve airflow.
- **CFTR modulators (e.g., ivacaftor, lumacaftor/ivacaftor, elexacaftor/tezacaftor/ivacaftor):** Newer drugs that target the defective CFTR protein. Some improve the function of misfolded CFTR protein at the cell surface; others help the protein fold correctly so it reaches the membrane. These represent a major breakthrough but only work for specific mutations.
- **Lung transplant:** May be necessary in severe cases when lung function deteriorates below a critical level.

Gene Therapy for CF

Note: gene therapy and the CF treatments above are not in the 9BN0 specification (Topic 2 covers the effects of the CF mutation and genetic screening). Treat this section as background.

Gene therapy aims to insert a functional copy of the CFTR gene into the epithelial cells of the lungs.

Approaches include:

- **Viral vectors (e.g., adenovirus, lentivirus):** Modified viruses deliver the gene efficiently, but may trigger an immune response, and the effect is temporary as epithelial cells are regularly replaced.
- **Liposome-mediated delivery:** The gene is wrapped in lipid vesicles (liposomes) that fuse with cell membranes. Safer than viral vectors but less efficient at gene delivery.
- **Challenges:** The gene must reach enough cells to make a clinical difference; the thick mucus barrier is difficult to penetrate; expression is temporary; repeated treatments may be needed; immune responses to vectors; and the gene must be expressed at the correct level.

Ethical considerations: Gene therapy for CF is somatic (affects only the patient, not their offspring), which is generally considered ethically acceptable. Germline gene therapy (which would be heritable) raises greater ethical concerns and is currently prohibited in humans in most countries.

Exam Tip: When explaining how CF affects gas exchange, link the molecular defect (CFTR chloride channel) to the cellular effect (thick mucus) to the tissue/organ level effect (blocked airways, infections) to the whole organism effect (breathing difficulty, reduced oxygen delivery). This shows you can explain at multiple levels of organisation.

2.6 Genetic Screening

- **Carrier testing:** Testing people with a family history of a genetic disorder (or their partners) to find out whether they carry a recessive allele. Two carriers then know they have a 1 in 4 chance of each child being affected, which helps them make decisions about having children.
- **Pre-implantation genetic diagnosis (PGD):** Embryos are produced by IVF. When an embryo is at about the 8-cell stage, one cell is removed and its DNA is tested, and only embryos without the disease-causing alleles are implanted. This avoids the decision about terminating a pregnancy. However, IVF is expensive with a low success rate, and unused embryos are destroyed.
- **Amniocentesis** (at about 15–20 weeks of pregnancy): Guided by ultrasound, a needle is inserted through the abdominal wall to take a sample of amniotic fluid, which contains fetal cells. The cells are tested for genetic disorders. There is a small risk of miscarriage (about 0.5–1%), and results can take a couple of weeks.
- **Chorionic villus sampling (CVS)** (at about 11–14 weeks): A small sample of the chorionic villi (part of the placenta, which has the same genotype as the fetus) is taken through the cervix or abdominal wall. It can be done earlier than amniocentesis, so any termination would be earlier, but it carries a similar or slightly higher risk of miscarriage.

Implications of prenatal screening: A positive result may lead parents to decide whether to terminate the pregnancy, or it gives them time to prepare to care for an affected child. The tests themselves carry a risk of miscarriage of a healthy fetus. False positives and false negatives can occur, and waiting for results and deciding what to do can cause great anxiety.

Social and Ethical Issues

Genetic screening can be discussed from a range of ethical viewpoints:

- **Rights and duties:** Some people believe there are absolute rules. For example, some hold that the embryo or fetus has a right to life, so termination, or discarding embryos in PGD, is always wrong.

Others stress parents' right to know about and plan for their child.

- **Maximising benefits and minimising harm (utilitarianism):** The right choice is the one giving the greatest good for the greatest number. This could mean weighing the suffering avoided, and the cost of lifelong care, against the harm of terminations and miscarriages caused by testing.
- **Autonomy (making decisions for yourself):** Individuals should have the information and freedom to make their own informed choices, without pressure.
- **Leading a virtuous life:** Acting as a virtuous person would, with compassion and integrity.

Social issues include the confidentiality of genetic information and possible discrimination (e.g. by employers or insurers), pressure on parents to terminate affected pregnancies, and concerns that screening could lead to "designer babies" or devalue people living with genetic conditions.